

Anti-Microbial

1. Preparation of rifamycin iminomethylenyl quinolone derivatives effective against drug-resistant microbes

Ding, C. Z., Jin, Y., Longgood, J. C., Ma, Z., Li, J., Kim, I. H., Minor, K. P. and Harran, S.

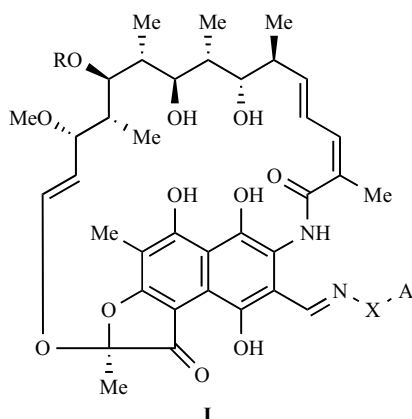
Cumbre Inc., 1502 Viceroy Drive, Dallas, TX 75235, USA

PCT Int. Appl. WO 2005, 70, 941 (Cl. C07D498/08), 4 Aug. 2005, US Appl. 2004/PV536, 018, 13 Jan. 2004; 117 pp; *C.A.* **143**(10): 172685q

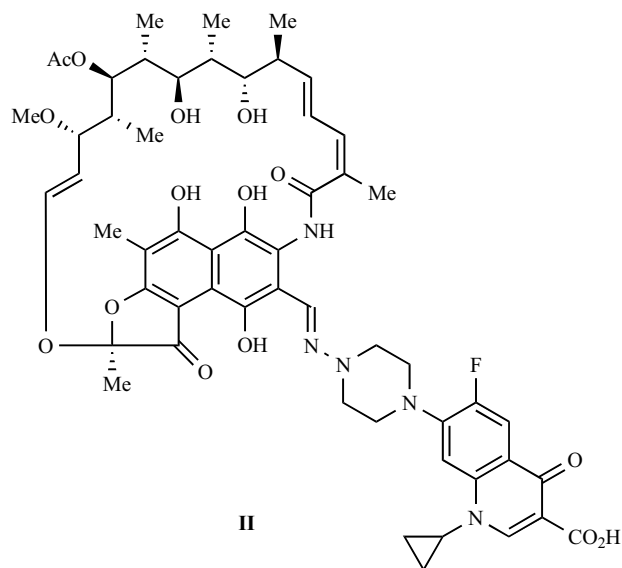
Abstract: This patent describes the preparation of rifamycin iminomethylenyl quinolone derivatives **I** and **II** along with their anti-microbial activities.

Activity: The synthetic compounds of the formula **I** showed activity against *Staphylococcus aureus* ATCC 29213 RpoB^{11418Y} with MIC = 0.06 - 16 µg/mL.

Origin: Synthetic



A = quinolone group; X = alkylene, arylene, heterocyclylene, CO, C=N, O, etc.; R = H, acetyl, etc.



Web URL: <http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?CMD=search&DB=pubmed>

2. Antifungal activity and studies on mode of action of novel xanthoxylene-derived chalcones

Boeck, P., Leal, P. C., Yunes, R. A., Cechinel, F. V. C., Lopez, S., Sortino, M., Escalante, A., Furlan, R. L.E. and Zacchino, S.

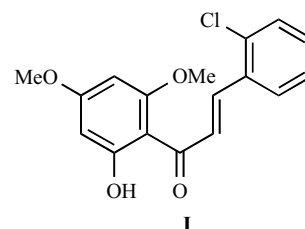
Departamento de Quimica, Universidade Federal de Santa Catarina, Florianopolis, Brazil

Archiv der Pharmazie 2005, **338**(2-3), 87-95; *C.A.* **143**(10): 169388q

Abstract: This manuscript highlights the antifungal properties of some novel synthetic xanthoxylene-derived chalcones.

Activity: The chalcone **I**, with a chlorine (Cl) atom at ortho position of the benzene ring B, exhibited best activity against the standard strains of *Trichophyton rubrum* with MIC₅₀ = 12.5 µg/mL.

Origin: Synthetic



Web URL: <http://www.mdpi.org/molecules/papers/91000815.pdf>

3. Synthesis of novel azo schiff bases and their antibacterial and antifungal activities

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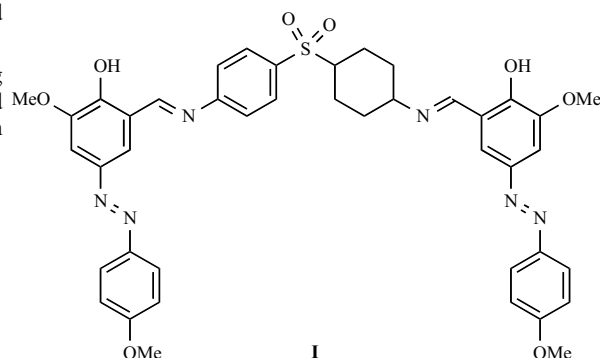
Department of Parasitology and Mycology, Shiraz Medical School, Shiraz University Medical Science, Shiraz 71345, Iran

Molecules 2004, 9(10), 815-824; *C.A.* **143**(11): 193756d

Abstract: Ten novel azo Schiff bases have been synthesized and evaluated as antifungal agents.

Activity: Of the series, compound **I** was found to exhibit promising activity against *Bacillus subtilis* (inhibition zone >12 mm) and moderate activity against *Staphylococcus aureus* (inhibition zone 9-12 mm).

Origin: Synthetic



Web URL: <http://www.sciencedirect.com/science/journal/02235234>

4. Synthesis and study of antibacterial and antifungal activities of novel 8-methyl-7,9-diaryl-1,2,4,8-tetraazaspiro[4.5]decan-3-thiones

Balasubramanian, S., Ramalingan, C., Aridoss, G. and Kabilan, S.

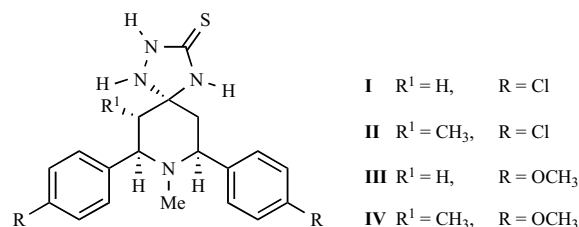
Department of Chemistry, Annamalai University, Annamalai Nagar, Tamil Nadu 608 002, India

European Journal of Medicinal Chemistry 2005, **40**(7), 694-700; *C.A.* **143**(11): 193956u

Abstract: Some novel 8-methyl-7,9-diaryl-1,2,4,8-tetraazaspiro[4.5]decan-3-thiones have been synthesized and evaluated for their antibacterial and antifungal activities.

Activity: Compounds, **I-IV** showed antibacterial activity against *Staphylococcus aureus* (NCLM-2492) (MIC = 25, 25, 50 and 25 $\mu\text{g/mL}$, respectively), against *Bacillus subtilis* (NCLM-2439) (MIC = 50, 25, 25 and 25 $\mu\text{g/mL}$, respectively), *Escherichia coli* (NCLM-2345) (MIC = 50, 25, 25 and 25 $\mu\text{g/mL}$, respectively), *Pseudomonas aeruginosa* (NCLM-2035) (MIC = 25, 25, 25 and 25 $\mu\text{g/mL}$, respectively).

Origin: Synthetic



5. Preparation of (aminohydroxyalkyl)quinolines as mycobacterial inhibitors

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PCT Int. Appl. WO 2005, **70**, 430 (Cl. A61K31/47), 4 Aug. 2005, US Appl. 2004/PV538, 907, 23 Jan. 2004; 88pp; *C.A.* **143**(11): 193916f

Abstract: This investigation describes the synthesis of (aminohydroxyalkyl)quinolines as mycobacterial inhibitors.

Activity: The title compound **I** exhibited $\text{pIC}_{50} = 6.6$ against *Mycobacterium smegmatis* ATCC607.

Origin: Synthetic

